

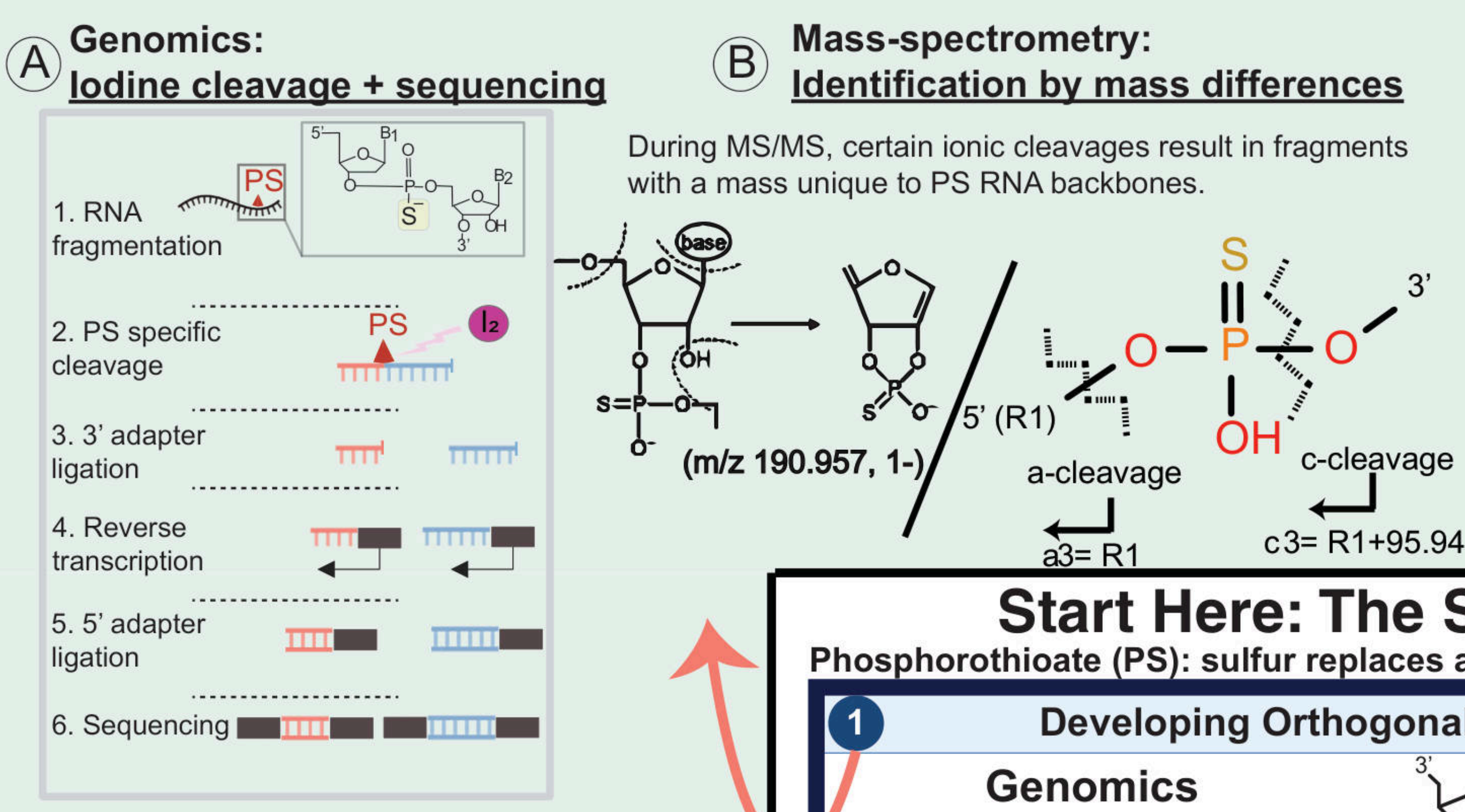
Discovery of RNA phosphorothioate modifications (PS) in anaerobic and thermophilic archaea: a dynamic backbone modification coupled to environmental conditions



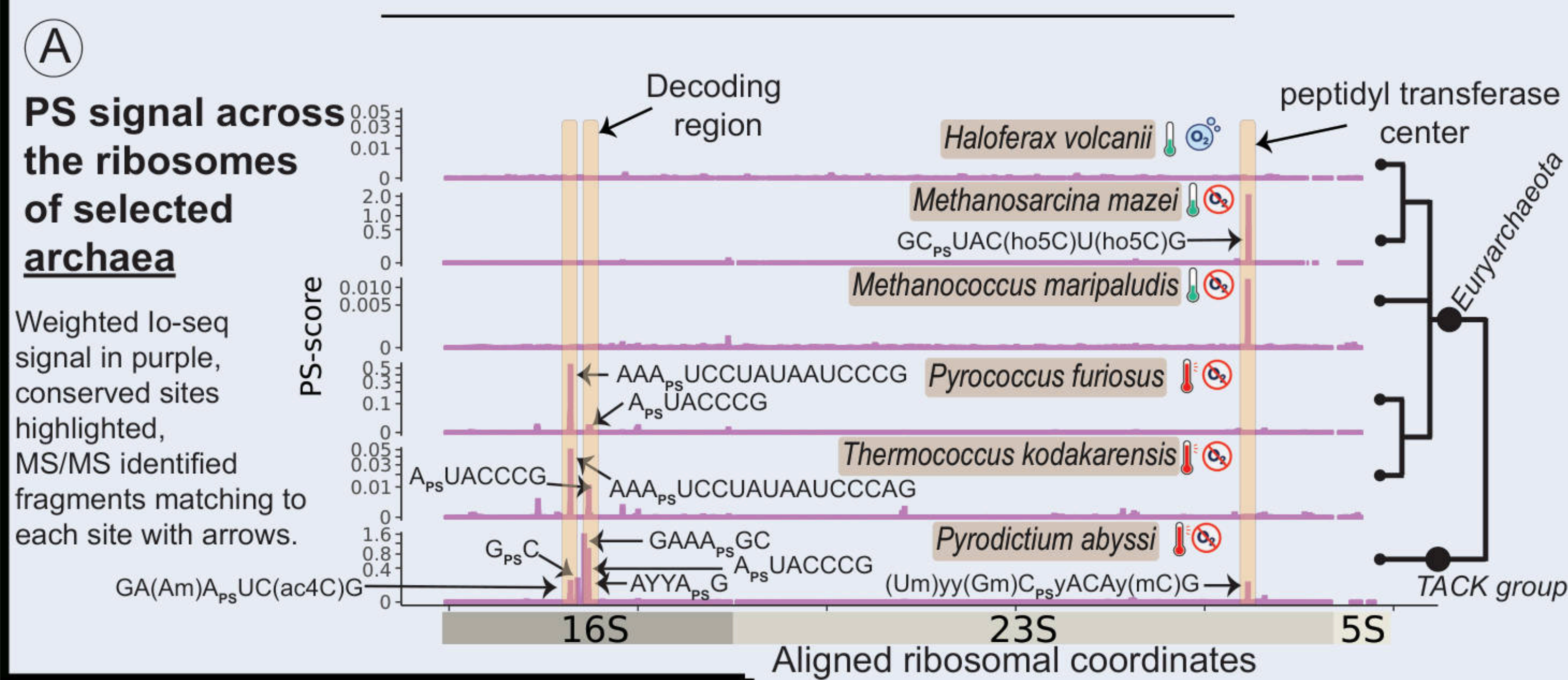
Alexander Maman¹, Yuko Nobe², Kristin A. Fluke³, Katrin Weidenbach⁴, Alexandra N. Harte³, Brett W. Burkhardt³, Jakub Nowak⁵, Priyadarshini Mukherjee⁶, Deepak Kumar Choudhary⁷, Anatoly Kustanovich¹, Ronit Nir¹, Katharina Vogl⁸, Kiall Suazo⁹, Danijela Radovanović¹⁰, Donna Matzov¹, Walter Rossmann¹⁰, Jordan L. Meier⁹, Moran Shalev-Benami¹, Dina Grohmann⁸, Kylie D. Allen⁶, Eric Westhof¹¹, Sebastian Glatt¹², Ruth A. Schmitz⁴, Thomas J. Santangelo^{3, #}, Masato Taoka^{2, #}, **Schrage Schwartz^{1, #}**

¹ Weizmann Institute of Science, Rehovot, Israel · ² Tokyo Metropolitan University, Tokyo, Japan · ³ Colorado State University, Fort Collins, CO, USA · ⁴ Kiel University, Kiel, Germany · ⁵ Jagiellonian University, Kraków, Poland · ⁶ Virginia Tech, Blacksburg, VA, USA · ⁷ Tel Aviv University, Tel Aviv, Israel · ⁸ University of Regensburg, Regensburg, Germany · ⁹ National Cancer Institute, Frederick, MD, USA · ¹⁰ Medical University of Vienna, Vienna, Austria · ¹¹ Université de Strasbourg/CNRS, Strasbourg, France · ¹² University of Veterinary Medicine Vienna, Vienna, Austria

1 Two orthogonal methods allow the detection and quantification of phosphorothioates at a single-nucleotide resolution

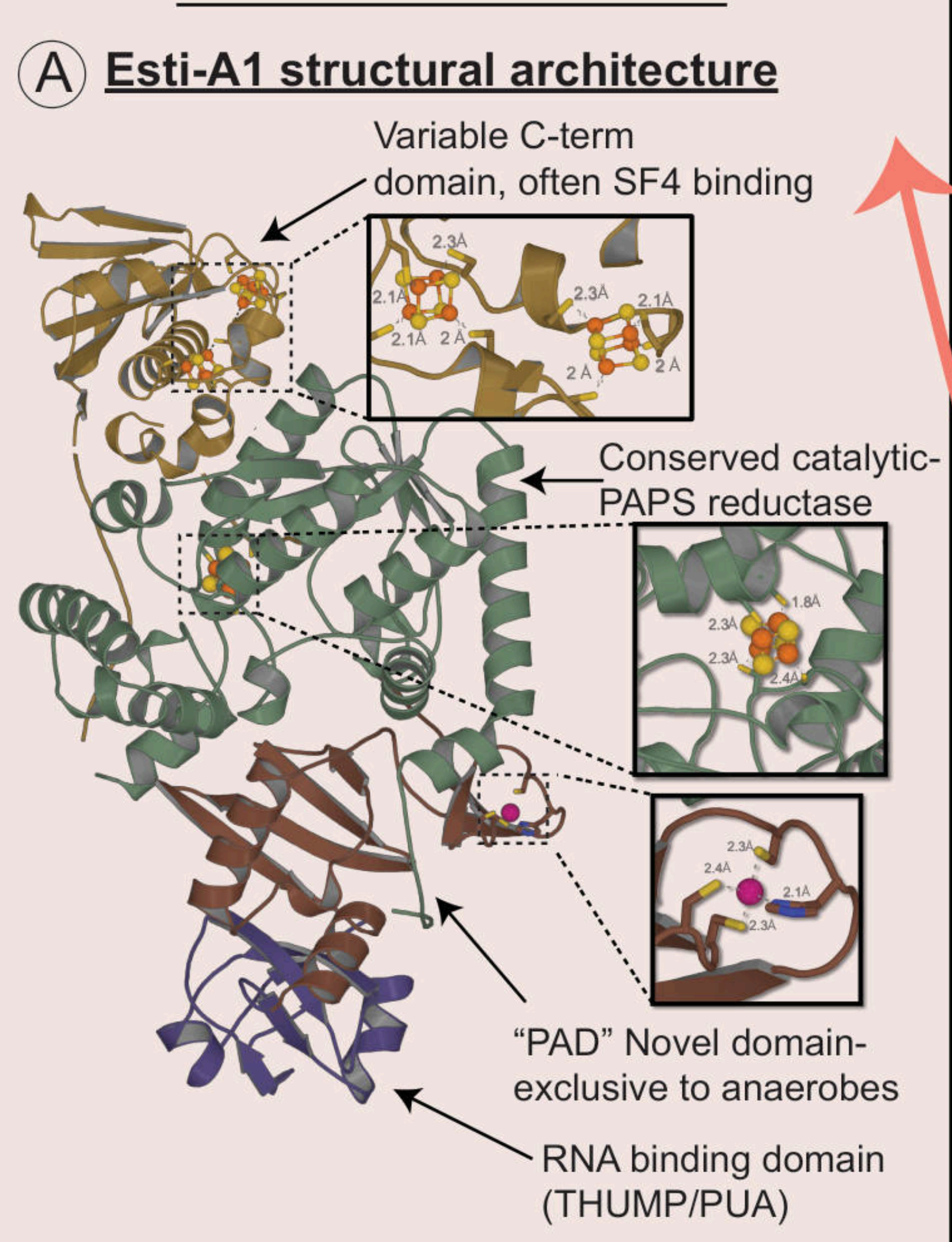


2 Mapping phosphorothioates across diverse archaea (A) reveals an environmentally dynamic mark (B,C), conserved at functional hotspots (A,D)

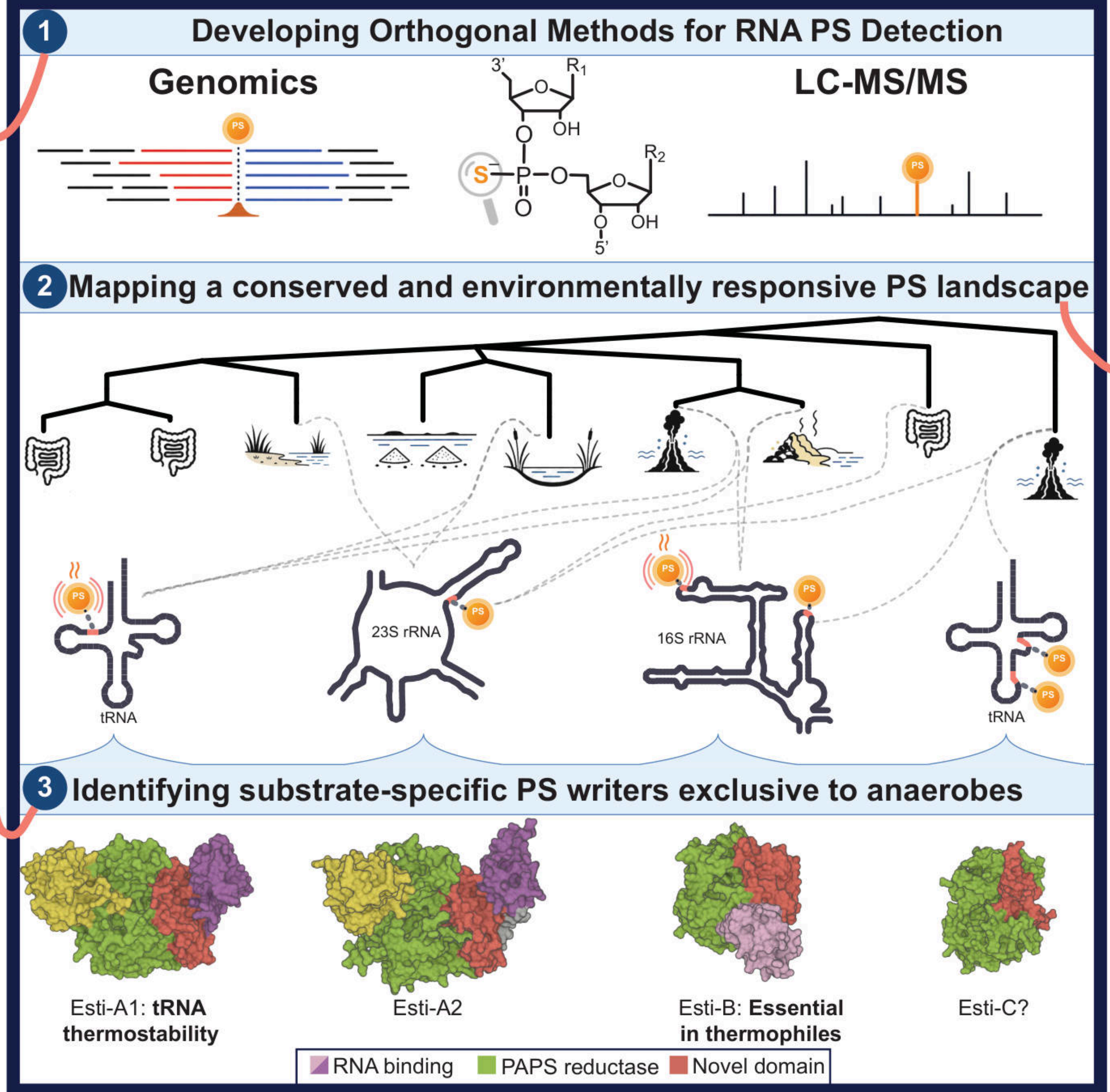


Digital poster & Cell paper
(Link will be updated upon publication)

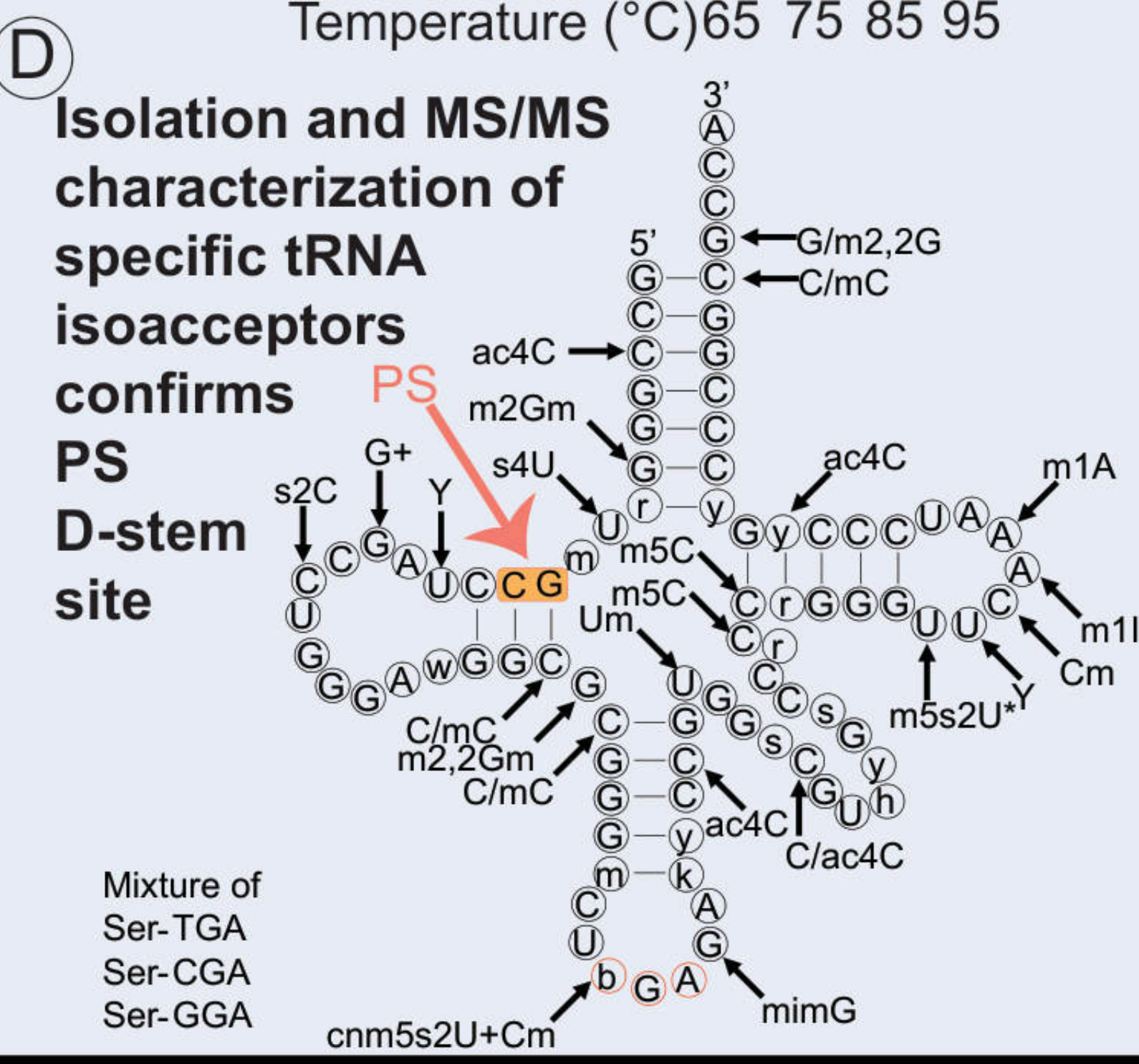
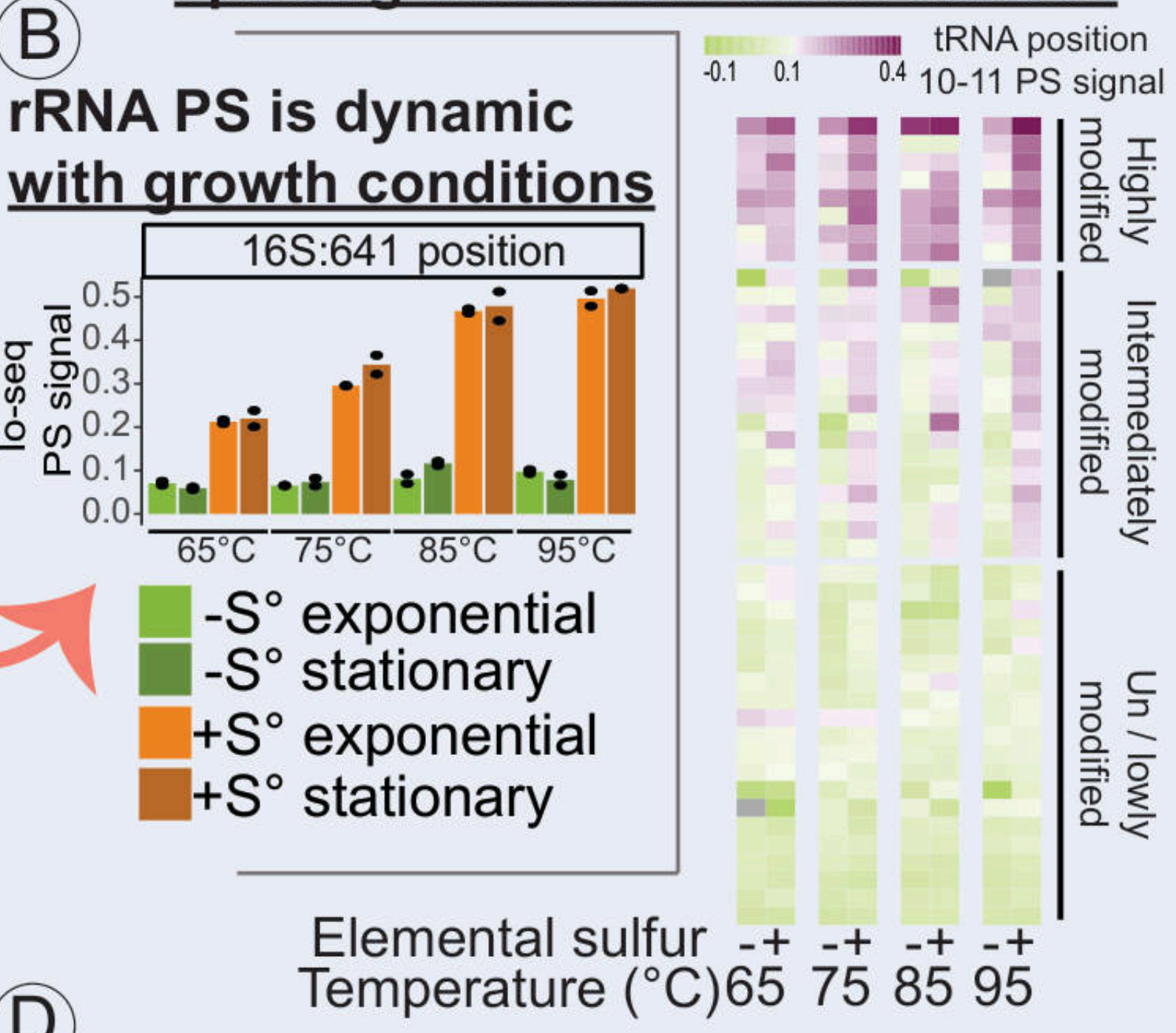
3 Discovery (A,B) and evolutionary characterization (C,D) of the depositing enzymes reveal a diverse (D) site-specific (E) family



Start Here: The Story at a Glance

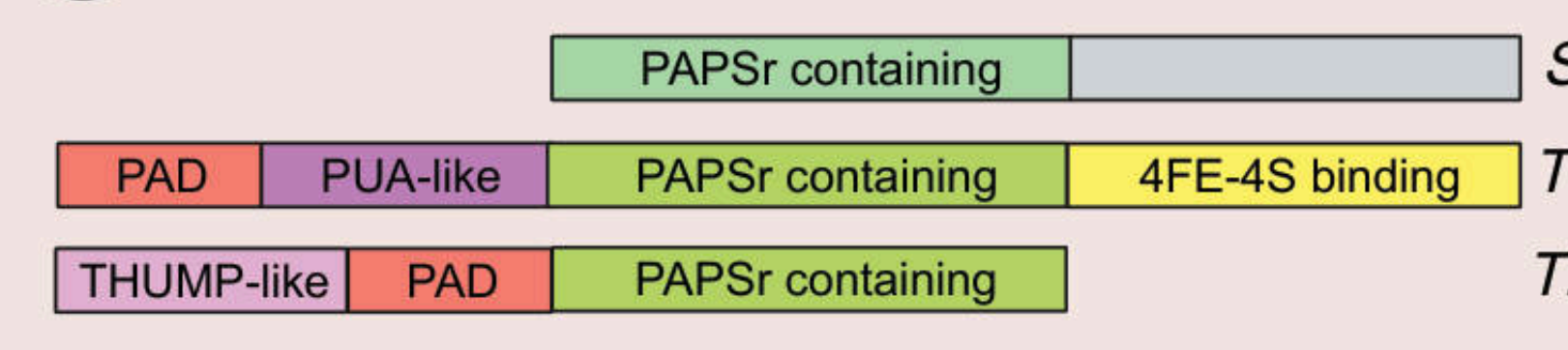


(C) A conserved D-stem tRNA PS site is dynamic with growth conditions, splitting tRNAs into three classes

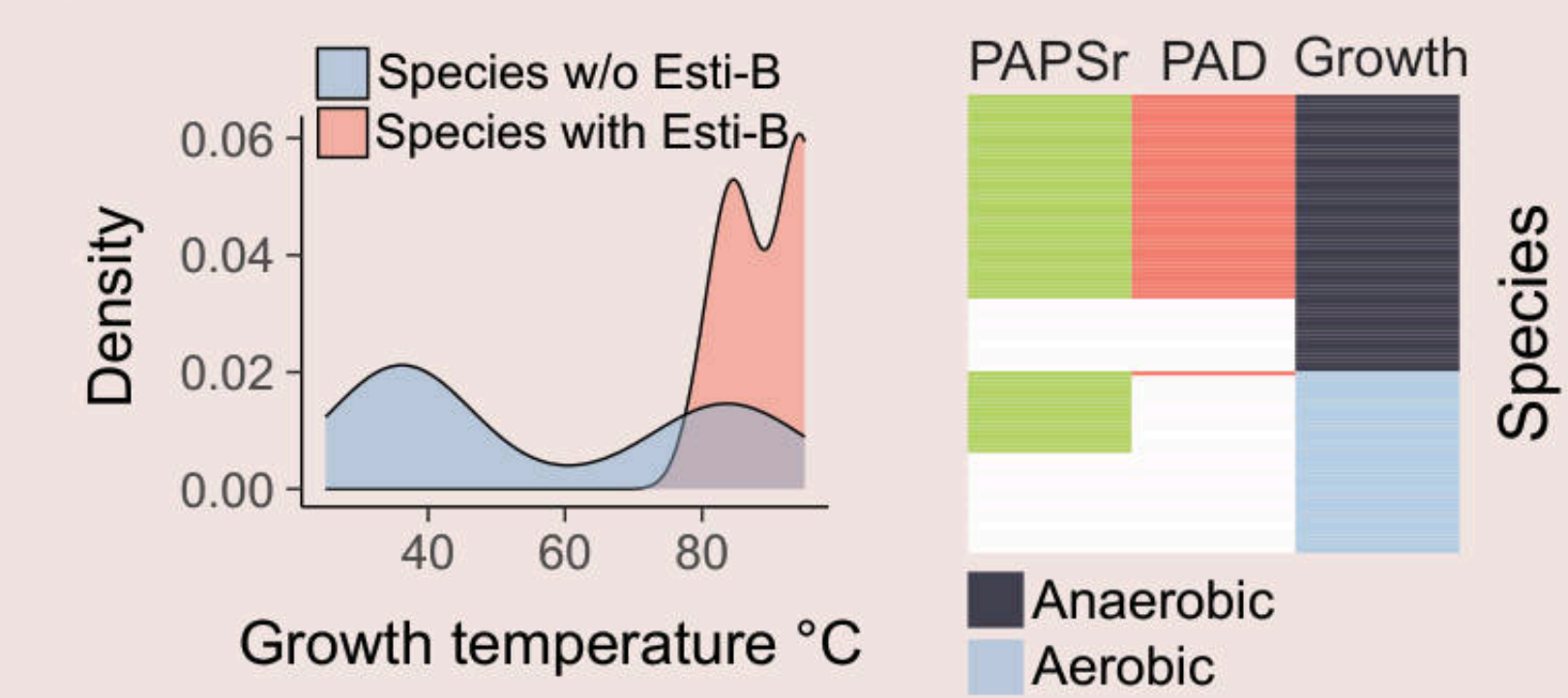


Conclusions & Outlook By comparing RNA modification landscapes across diverse organisms, we discovered phosphorothioates (PS) as natural modifications of the RNA phosphate backbone, exclusive in our survey to anaerobic archaea. Developing bespoke sequencing methods, we map PS to conserved rRNA and tRNA hotspots, while genetic analyses identified their **Esti writer enzymes**. PS levels **vary with temperature and sulfur availability** and contribute to RNA stability and **thermoadaptation**. The work illustrates how **cross-species comparative profiling can uncover previously hidden biology**, and suggests that environmentally responsive RNA modifications such as PS could provide molecular **readouts of the physiological state of populations**.

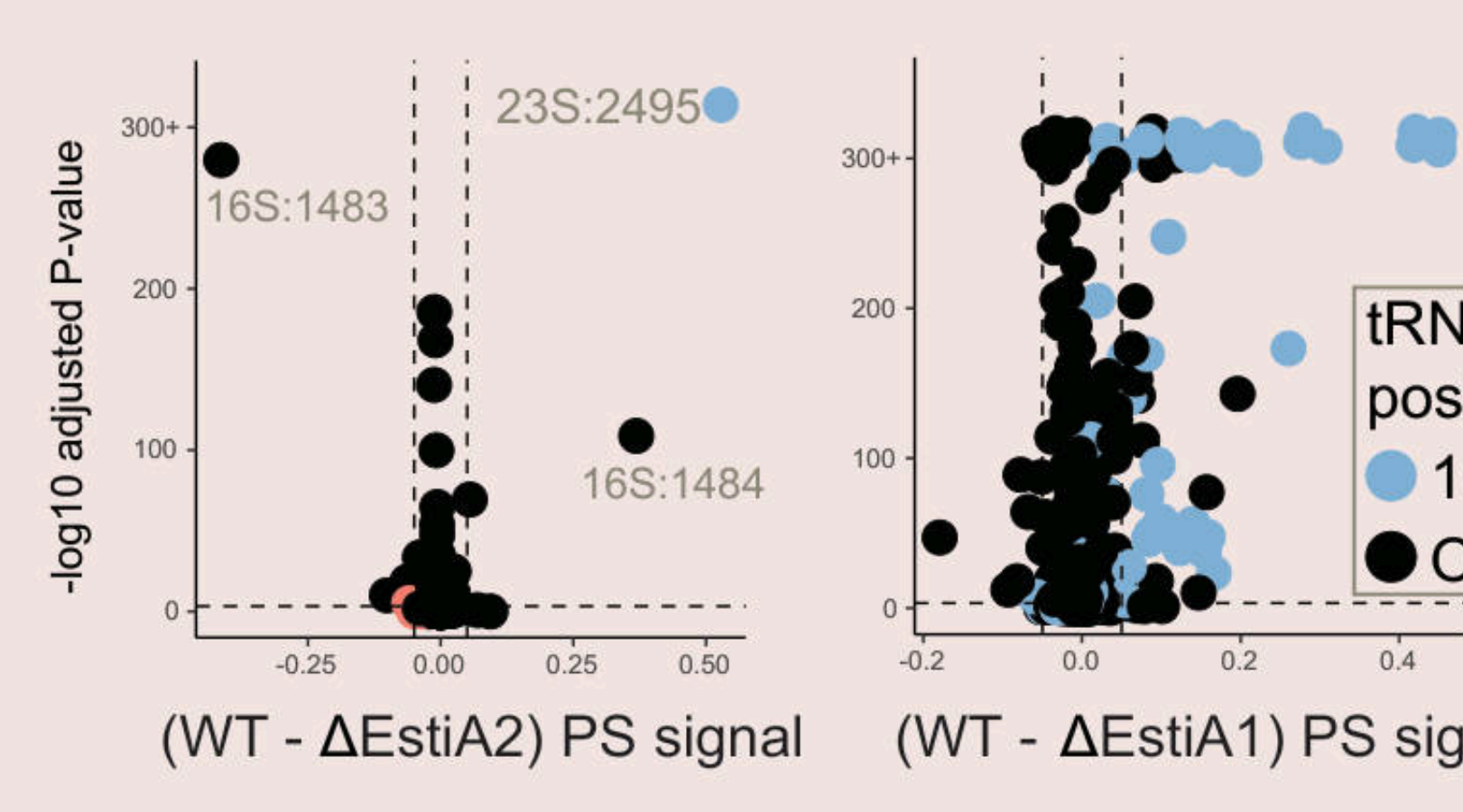
(B) DNA vs RNA modifying architecture



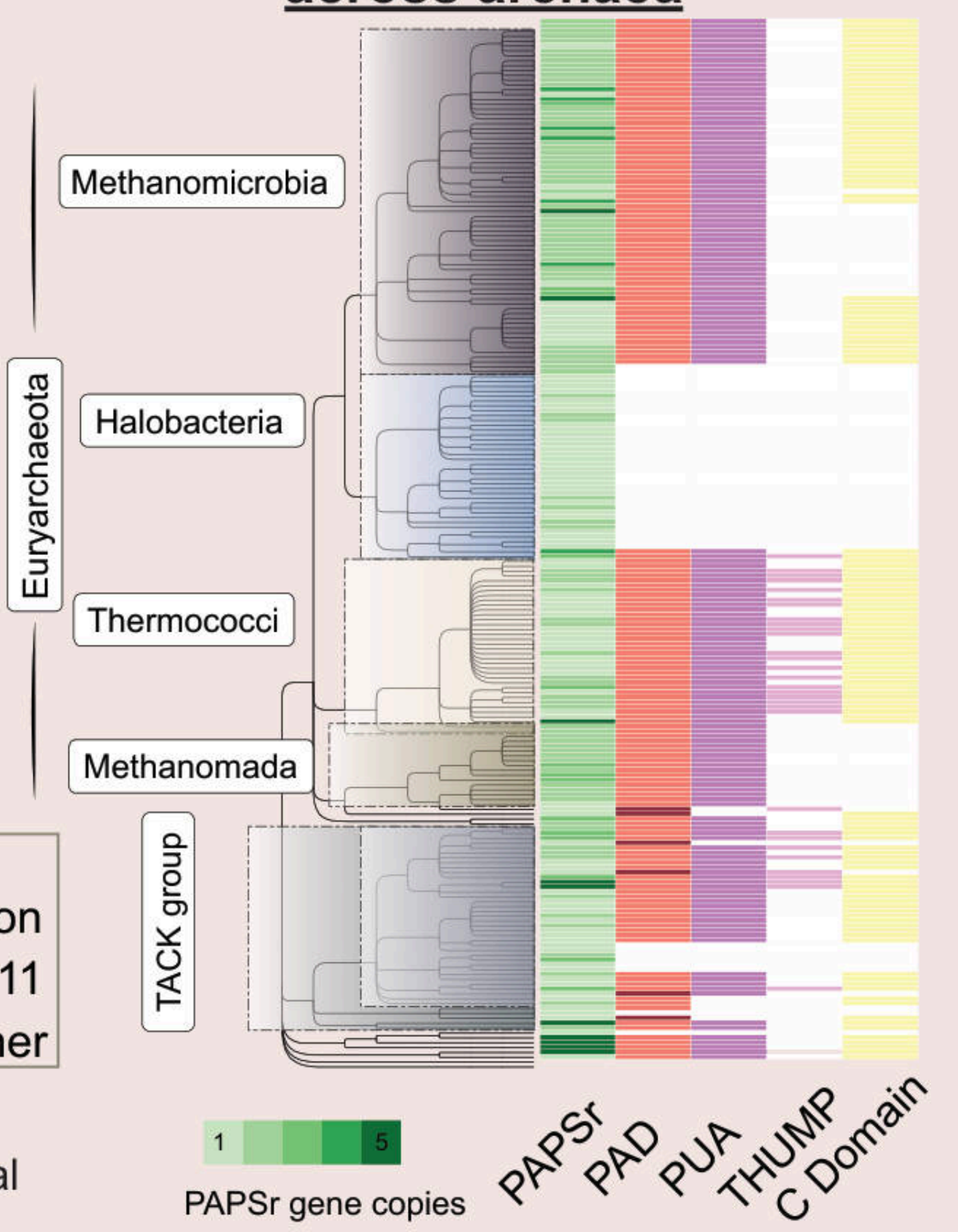
(C) Variants track specific ecological niches



(E) Esti-A1 & Esti-A2 mutants in *M. mazeri* show enzyme-site specificity



(D) Diverse architecture across archaea



T. kodakarensis tRNA modifying Esti-A1 mutants have temperature related physiological (A) and molecular phenotypes (B, C). And PS localization around functional regions of interest (D, E)

